

HR WORKFORCE & ATTRITION ANALYSIS

PROJECT REPORT

Human Resources (HR) Analytics
Employee Attrition and Retention Analysis

1. Project Overview and Objective

This project focuses on analysing employee data to understand workforce size, employee attrition, department and job-role performance, and some factors related to employee retention. The project starts with checking and cleaning the HR dataset using Python and then presents the analysis through an interactive Power BI dashboard.

- Analyse the total number of employees and employees who left the company.
- Find the departments and job roles with higher attrition.
- Study attrition based on age, gender and marital status.
- Compare attrition between employees who work overtime and those who do not.
- Study the relationship between job satisfaction and attrition.
- Understand work-life balance and employee tenure in relation to attrition.
- Create an interactive Power BI dashboard and give simple HR recommendations.

2. Data Sources

Item	Description
Domain	Human Resources / HR Analytics
Dataset	HR Employee Dataset
Records	1,470 employees
Main Target	Attrition
Dashboard Pages	2
Tools Used	Python, Pandas, NumPy, Matplotlib / Seaborn, Google Colab and Power BI

3. Problem Statement

- To understand how many employees are leaving the company.
- To identify departments and job roles where attrition is higher.
- To understand whether overtime is related to employee attrition.
- To study whether job satisfaction and work-life balance are related to attrition.
- To identify employee groups that may need more attention from HR.
- To provide simple recommendations based on the dashboard results.

4. Attribute (Column / Features) Details

Attribute Name	Data Type	Description
Age	Numeric	Age of the employee.
Attrition	Categorical	Shows whether the employee left the company.
BusinessTravel	Categorical	Employee business travel frequency.
Department	Categorical	Department of the employee.
DistanceFromHome	Numeric	Distance from home to workplace.
Education	Numeric	Education level.
EducationField	Categorical	Employee education field.
EnvironmentSatisfaction	Numeric	Satisfaction with work environment.
Gender	Categorical	Employee gender.
JobInvolvement	Numeric	Level of job involvement.
JobLevel	Numeric	Employee job level.

JobRole	Categorical	Employee job role.
JobSatisfaction	Numeric	Employee job satisfaction rating.
MaritalStatus	Categorical	Employee marital status.
MonthlyIncome	Numeric	Monthly employee income.
NumCompaniesWorked	Numeric	Number of companies previously worked for.
OverTime	Categorical	Whether the employee works overtime.
PerformanceRating	Numeric	Employee performance rating.
RelationshipSatisfaction	Numeric	Workplace relationship satisfaction.
TotalWorkingYears	Numeric	Total years of work experience.
TrainingTimesLastYear	Numeric	Training completed last year.
WorkLifeBalance	Numeric	Work-life balance rating.
YearsAtCompany	Numeric	Years worked at the current company.
YearsInCurrentRole	Numeric	Years in the current role.
YearsSinceLastPromotion	Numeric	Years since last promotion.
YearsWithCurrManager	Numeric	Years with the current manager.

5. Tools & Technologies

Tool	Use in Project
Python	Data loading, cleaning and analysis.
Pandas	DataFrame operations, missing values and grouping.
NumPy	Numerical operations.
Matplotlib / Seaborn	Charts and data visualization.
Google Colab	Running the Python notebook.
Power BI	Interactive dashboard and report.
DAX	KPI and calculated measures.

6. Data Pre-Processing (Python)

Python was used to check and clean the HR dataset before using the data for analysis and Power BI reporting.

6.1 Initial Data Quality Check

- Dataset contains 1,470 employee records.
- Checked rows, columns, data types and first few records.
- Checked missing values and duplicate rows.
- Reviewed categorical values and invalid numerical values.

6.2 Missing Value Handling

Numerical missing values were handled using median values. Categorical missing values were handled using the mode where appropriate. Missing Attrition values were not guessed and those records were removed.

6.3 Categorical Standardization

Text values were cleaned by removing extra spaces and standardizing common values such as Yes/No, department names, job roles and other categories.

6.4 Invalid Values and Duplicates

Invalid negative values in fields where negative values do not make sense were treated as missing values and then handled. Duplicate rows were checked and removed.

6.5 Final Clean Dataset

The cleaned dataset was saved separately from the unclean dataset so that the original data was not changed.

Missing values Handling:

	0
Age	73
Attrition	58
BusinessTravel	58
DailyRate	73
Department	58
DistanceFromHome	73
Education	0
EducationField	58
EmployeeCount	0
EmployeeNumber	0
EnvironmentSatisfaction	0
Gender	58
HourlyRate	73
JobInvolvement	0
JobLevel	0
JobRole	58
JobSatisfaction	0
MaritalStatus	58

	0
Age	0
Attrition	0
DailyRate	0
Department	0
DistanceFromHome	0
Education	0
EducationField	0
EmployeeNumber	0
EnvironmentSatisfaction	0
Gender	0
HourlyRate	0
JobInvolvement	0
JobLevel	0
JobRole	0
JobSatisfaction	0
MaritalStatus	0
MonthlyIncome	0
MonthlyRate	0

Datatype Handling:

	0		0
Age	int64	Age	float64
Attrition	object	Attrition	object
DailyRate	int64	BusinessTravel	object
Department	object	DailyRate	float64
DistanceFromHome	int64	Department	object
Education	int64	DistanceFromHome	float64
EducationField	object	Education	int64
EmployeeNumber	int64	EducationField	object
EnvironmentSatisfaction	int64	EmployeeCount	int64
Gender	object	EmployeeNumber	int64
HourlyRate	int64	EnvironmentSatisfaction	int64
JobInvolvement	int64	Gender	object
JobLevel	int64	HourlyRate	float64
JobRole	object	JobInvolvement	int64
JobSatisfaction	int64	JobLevel	int64
MaritalStatus	object	JobRole	object
MonthlyIncome	int64	JobSatisfaction	int64
MonthlyRate	int64	MaritalStatus	object
NumCompaniesWorked	int64	MonthlyIncome	float64
OverTime	object	MonthlyRate	float64
PercentSalaryHike	int64	NumCompaniesWorked	int64
PerformanceRating	int64	Over18	object
RelationshipSatisfaction	int64	OverTime	object
StockOptionLevel	int64	PercentSalaryHike	int64
TotalWorkingYears	int64	PerformanceRating	int64
TrainingTimesLastYear	int64	RelationshipSatisfaction	int64

7. Data Modelling and DAX (Power BI)

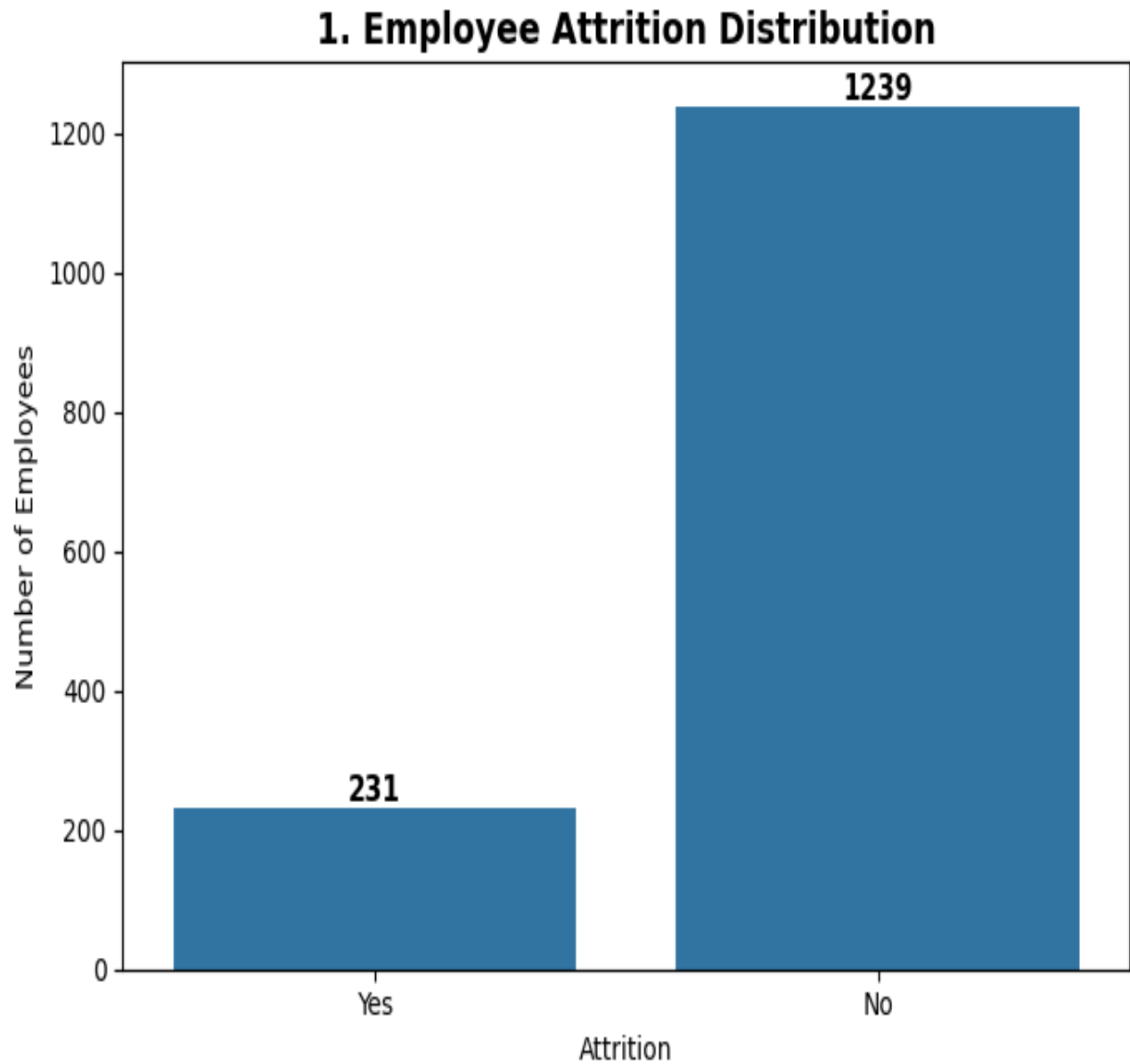
The cleaned HR dataset was loaded into Power BI for dashboard creation. The dashboard uses employee-level attributes and calculated measures to show attrition and workforce information.

Dashboard Metric / Area	Purpose
Total Employees	Shows the total number of employee records.
Total Employees Left	Shows employees whose Attrition value is Yes.
Attrition Rate	Shows the percentage of employees who left.
Average Monthly Income	Shows the average monthly income.
Average Job Satisfaction	Shows the average job satisfaction rating.
Average Environment Satisfaction	Shows the average environment satisfaction rating.
Average Tenure	Shows average years of service.
Overtime Attrition Rate	Shows attrition for overtime employees.

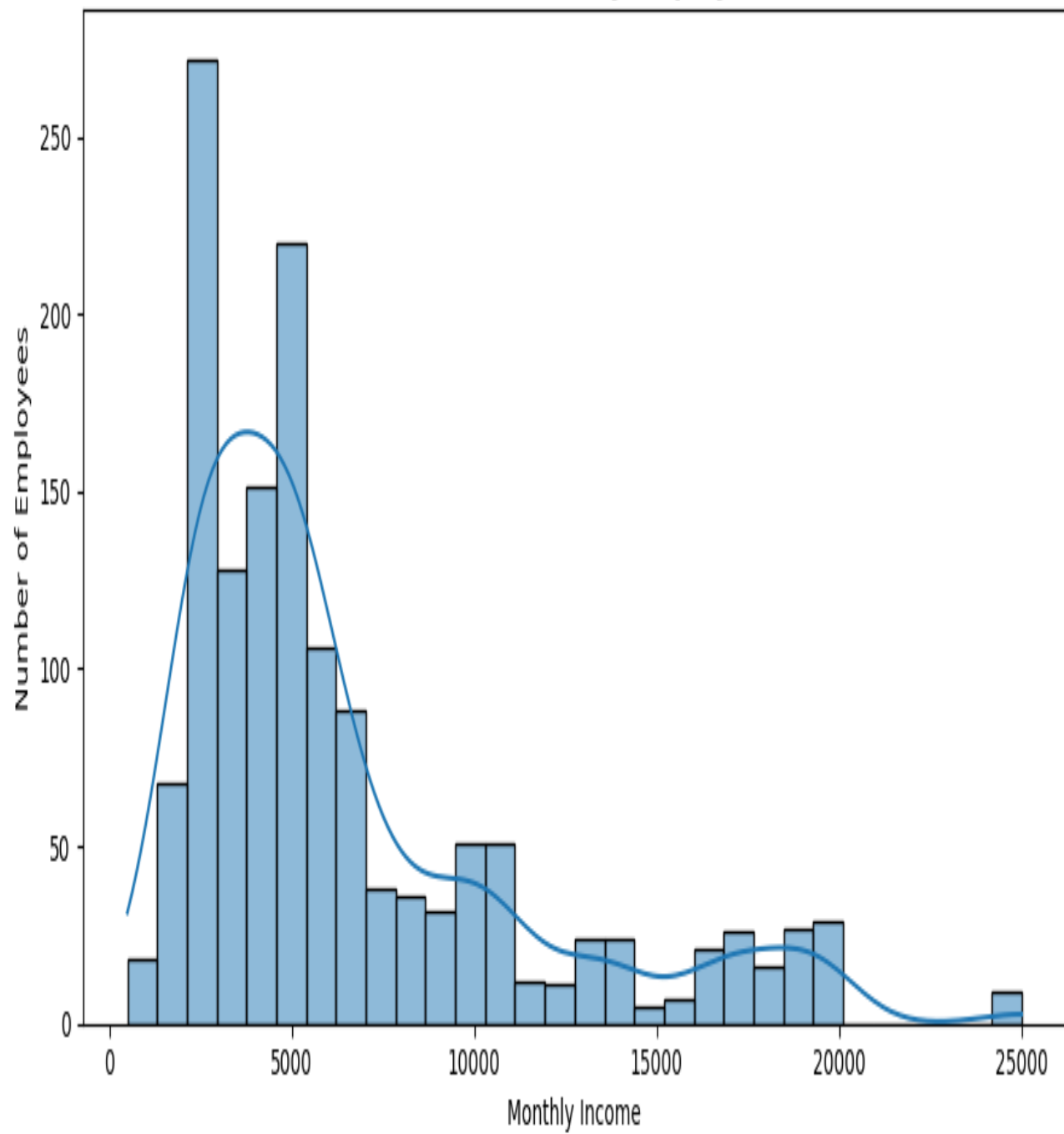
8. Python Visualizations

The following visualizations were created using Python in Google Colab to understand the HR employee data before creating the Power BI dashboard.

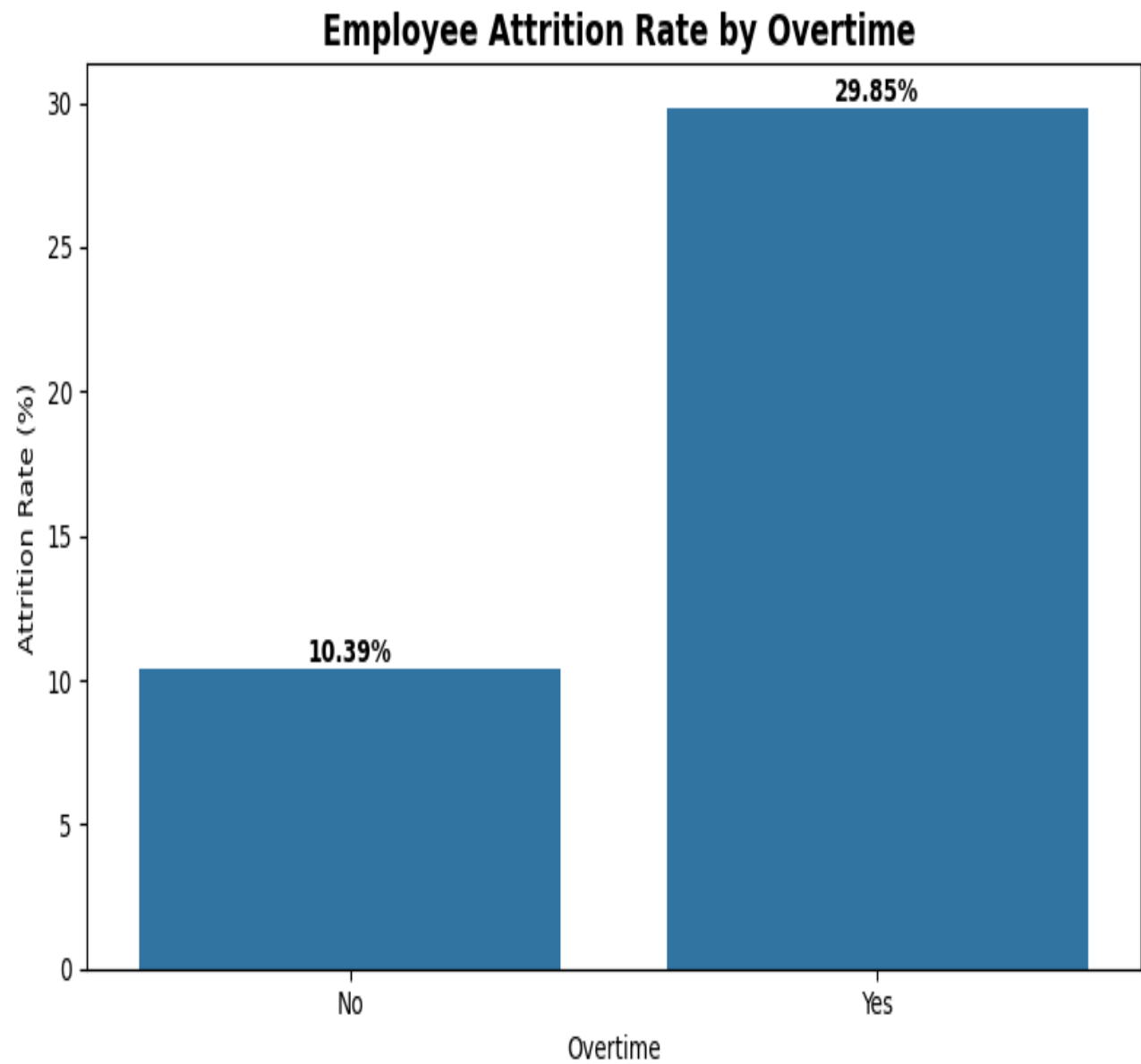
8.1 Employee Attrition Distribution



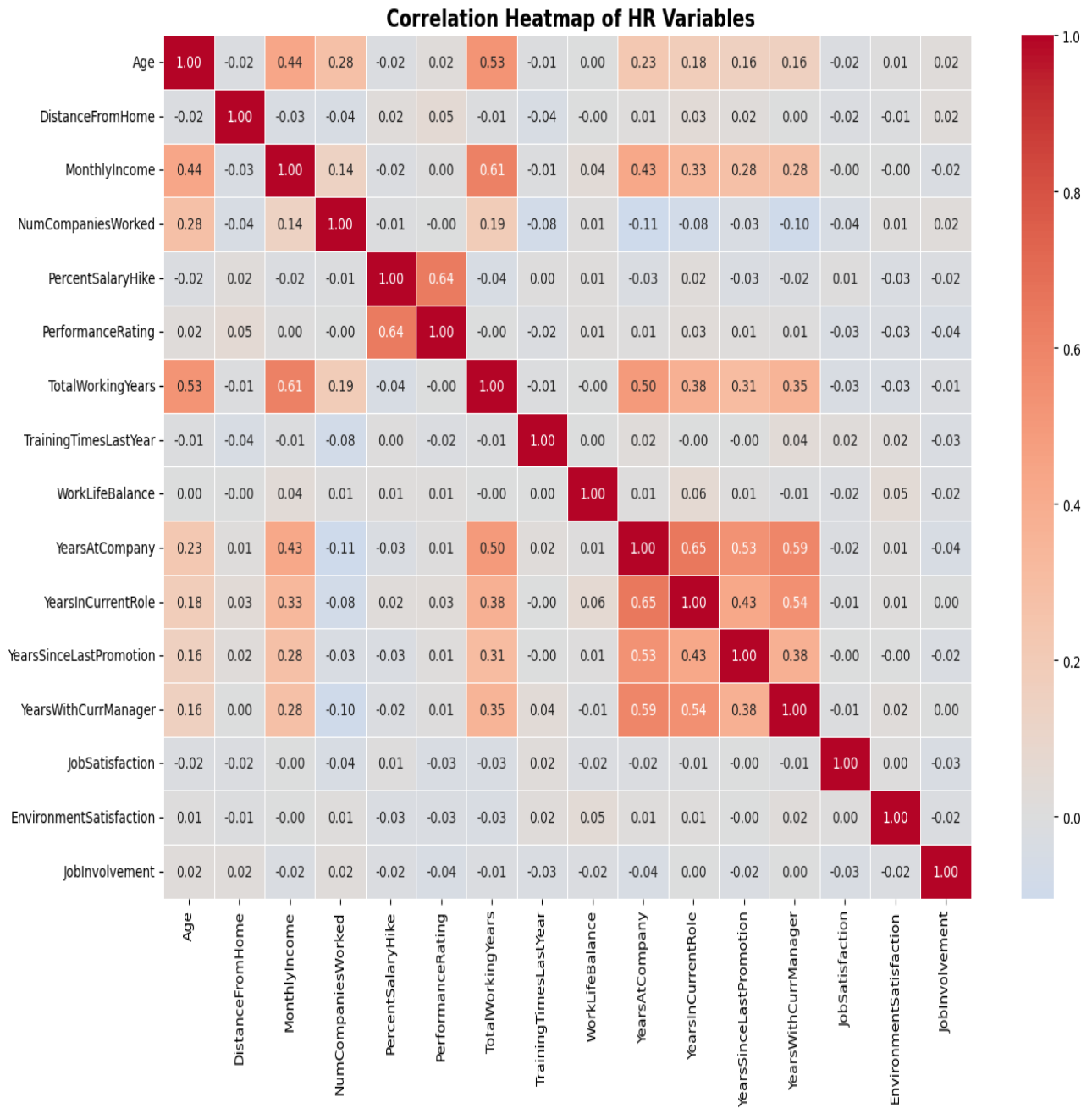
2. Distribution of Monthly Employee Income



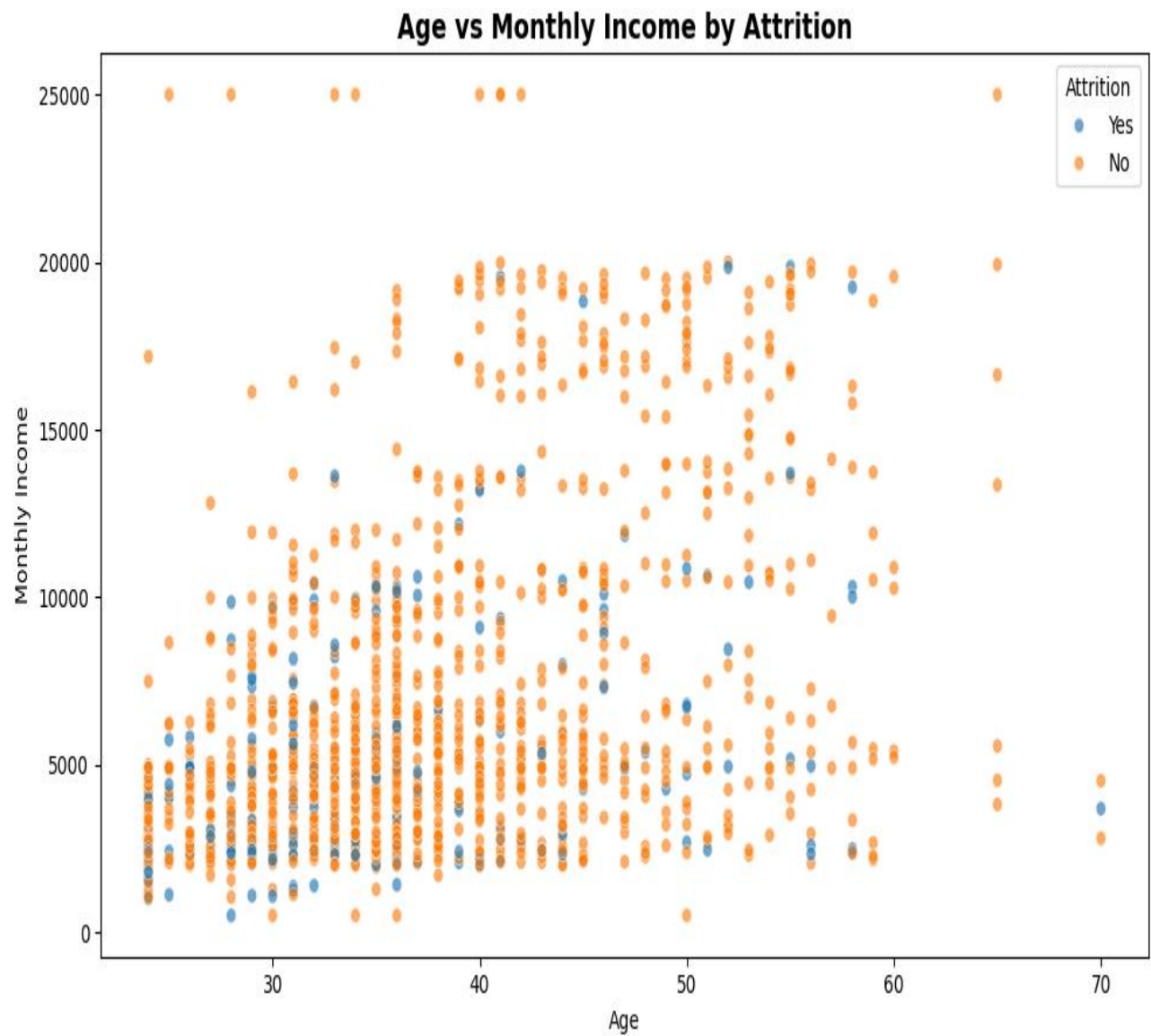
8.3 Employee Attrition Rate by Overtime



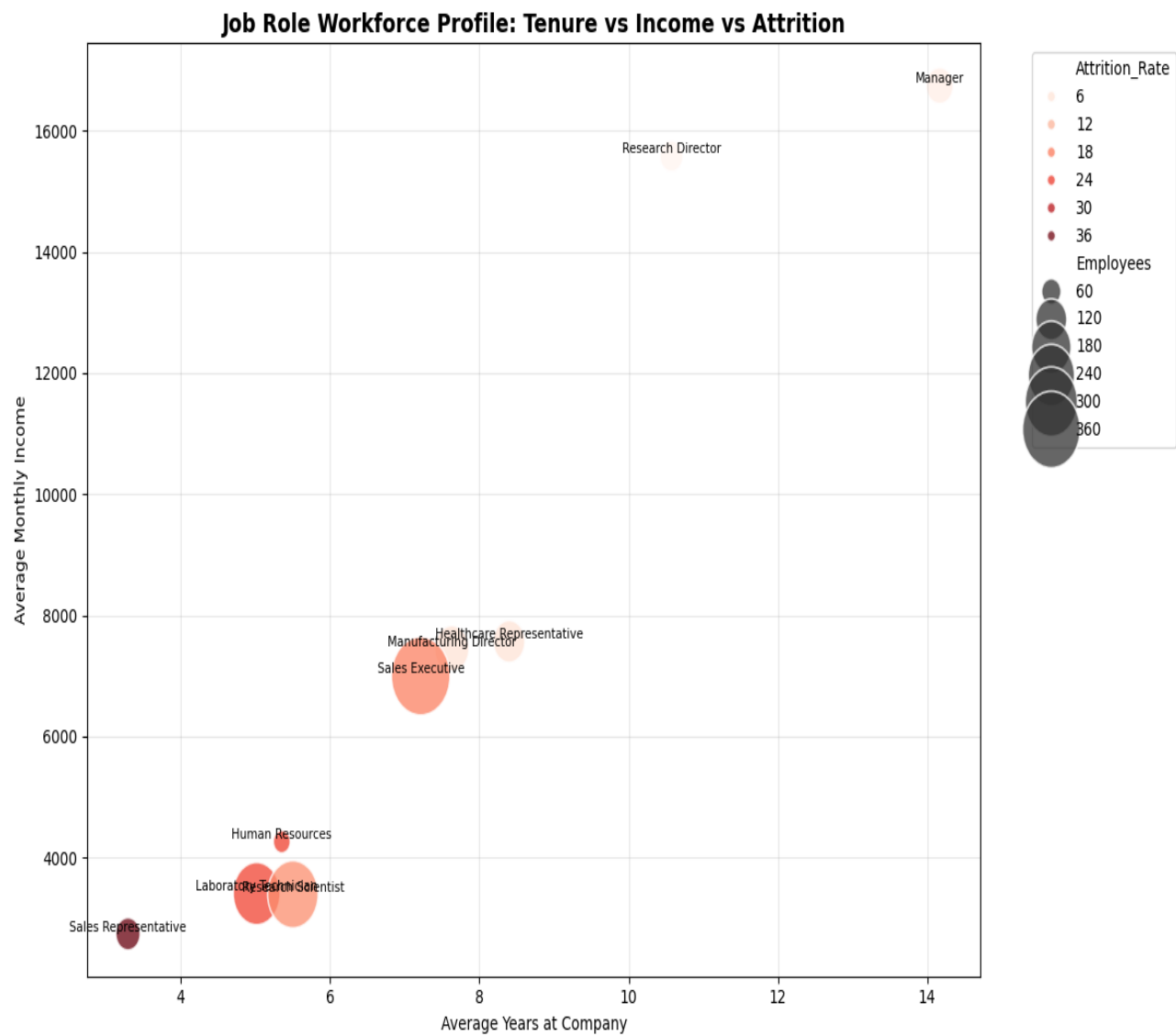
8.4 Correlation Heatmap of HR Variables



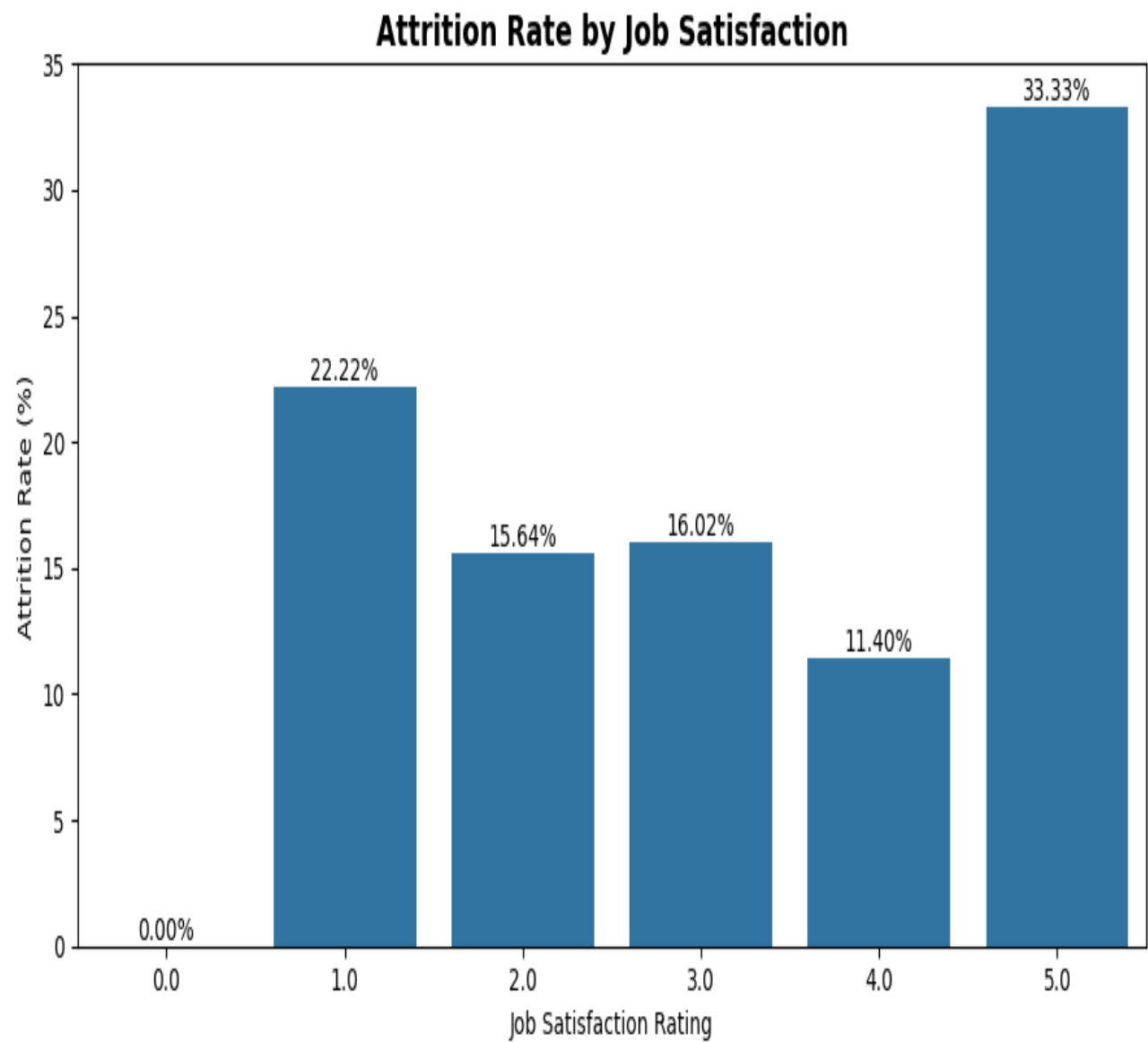
8.5 Age vs Monthly Income by Attrition



8.6 Job Role Workforce Profile: Tenure vs Income vs Attrition

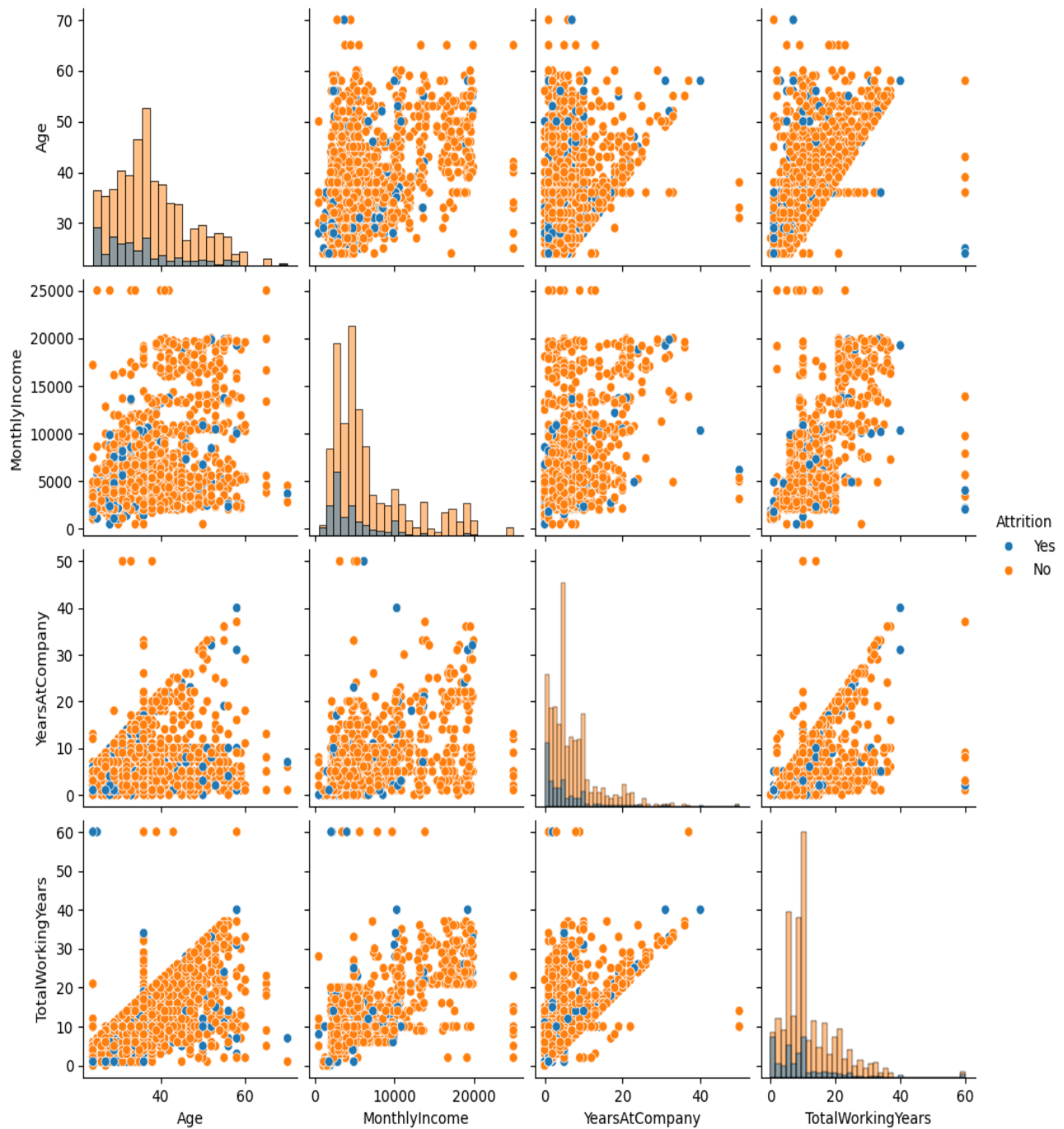


8.7 Attrition Rate by Job Satisfaction

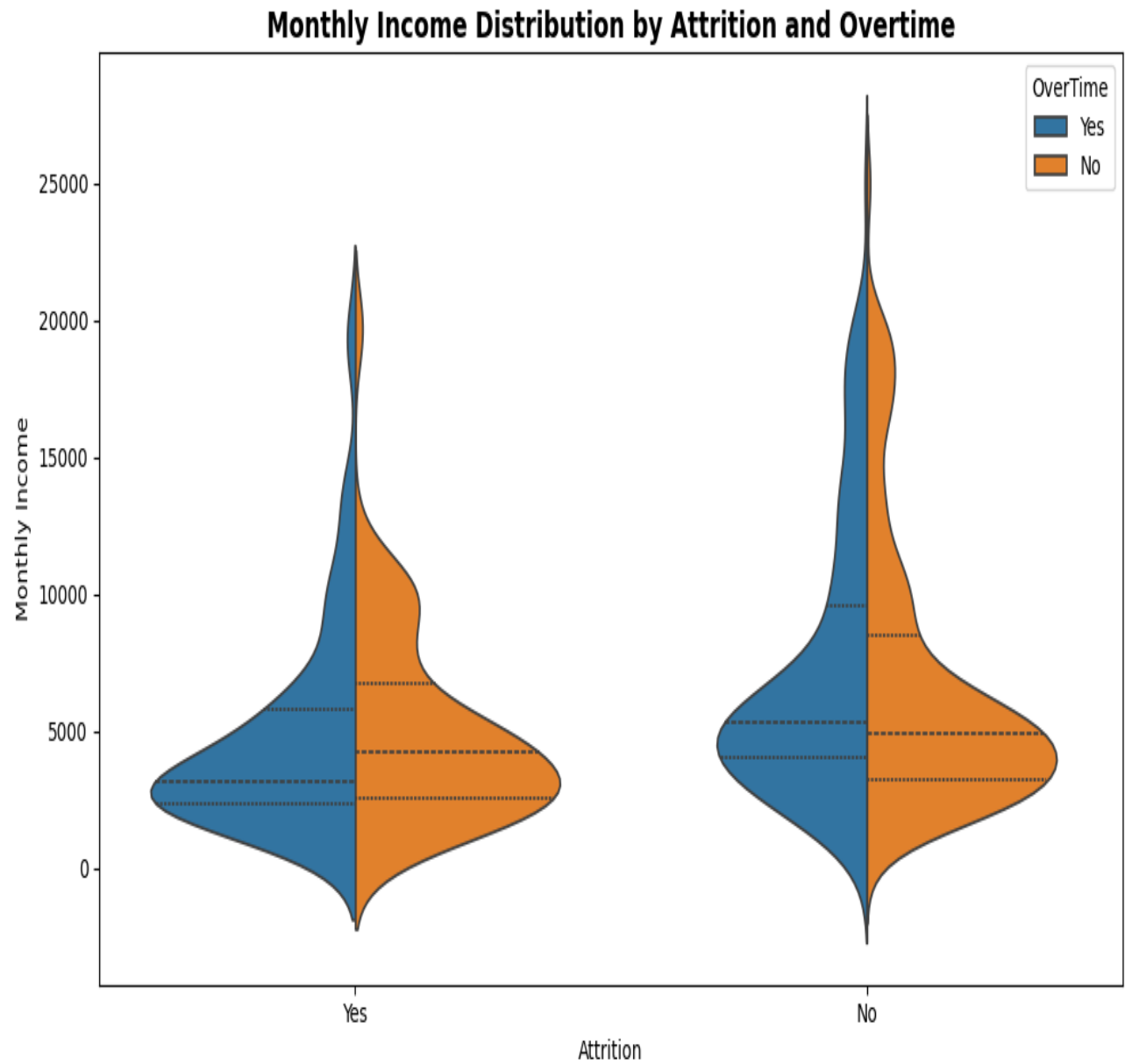


8.8 Pairplot – Multivariate Relationship Among HR Variables

1. Pairplot - Multivariate Relationship Among HR Variables



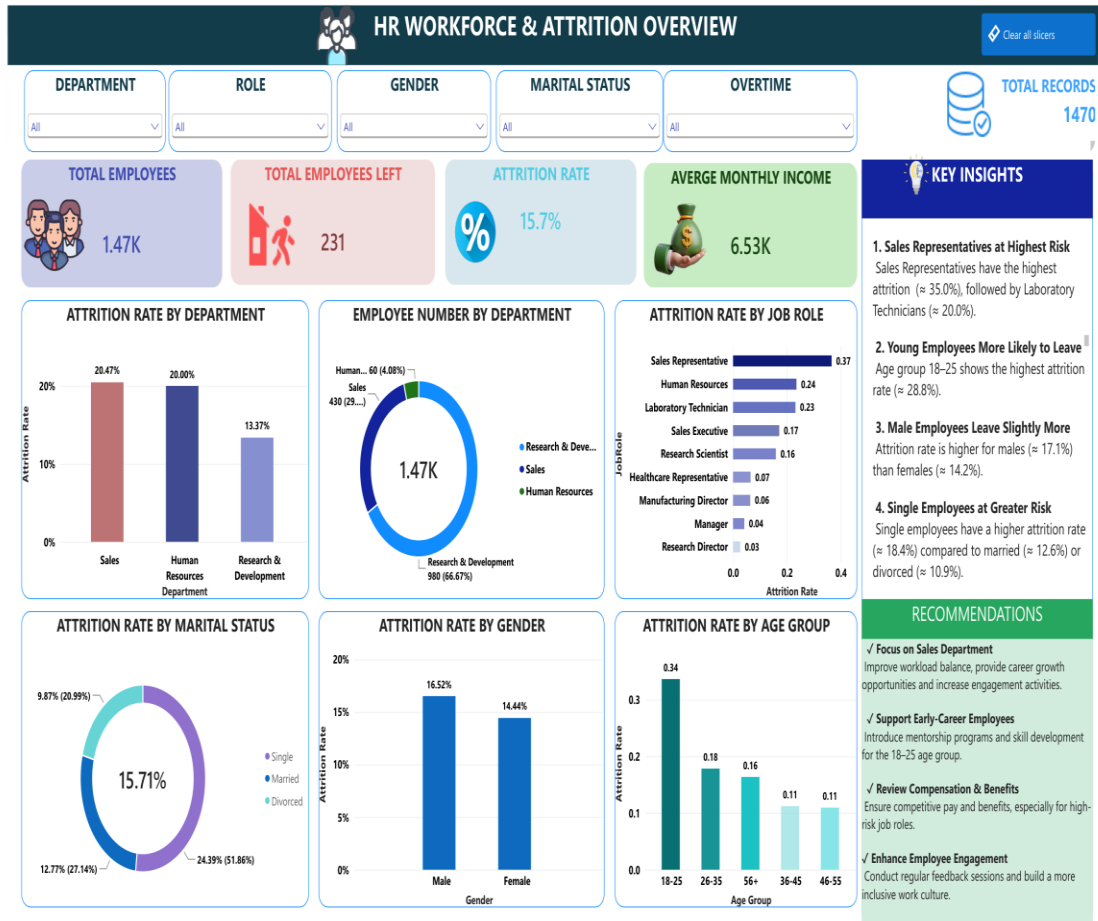
8.9 Monthly Income Distribution by Attrition and Overtime



9. Analysis and Visualizations (Power BI)

9.1 Power BI Dashboard – HR Workforce & Attrition Overview

The first dashboard gives an overall view of the workforce. It contains slicers for Department, Role, Gender, Marital Status and Overtime. It also contains KPI cards and charts for department, job role, age group, gender and marital status.



9.2 Power BI Dashboard – Employee Engagement & Retention Analysis

The second dashboard focuses on employee engagement and retention. It shows job satisfaction, environment satisfaction, tenure, overtime attrition, job level, work-life balance and income.



10. Insights & Recommendations

10.1 Key Insights

- The dashboard contains 1,470 employee records and shows 231 employees left.
- The overall attrition rate shown in the first dashboard is 15.7%.
- Average monthly income is shown as 6.53K.
- Sales and Human Resources show higher department attrition than Research & Development in the dashboard.
- Sales Representatives have the highest attrition among the displayed job roles.
- The 18–25 age group has the highest displayed attrition among the age groups.
- Male employees show slightly higher attrition than female employees.
- Single employees show higher attrition than married and divorced employees.
- Employees who work overtime show much higher attrition than employees who do not work overtime.
- The second dashboard shows average job satisfaction of 2.72 and average environment satisfaction of 2.73.
- Average tenure is shown as 7 years.
- The dashboard shows higher attrition for employees with weaker work-life balance.
- Employees with shorter tenure are shown as a higher-risk group.

10.2 Descriptive Insights

- Sales Representatives have the highest displayed attrition among the listed job roles.
- The 18–25 age group has the highest displayed attrition among the shown age groups.
- Single employees show higher attrition than married and divorced employees.
- Overtime employees show higher attrition than non-overtime employees.
- Short-tenure employees show higher attrition in the dashboard.

10.3 Diagnostic Insights

- Overtime is one of the clearest differences shown in the dashboard.
- Attrition changes across job roles and departments, so the same HR action may not work equally for every group.
- Lower job satisfaction and poorer work-life balance are linked with higher displayed attrition rates.
- Early-career and short-tenure employees need more attention from HR.

10.4 Predictive Insights

- Employees with high overtime, short tenure or lower satisfaction can be considered groups for closer retention monitoring.
- Regularly checking attrition by role, tenure and engagement can help HR find changes early.

10.5 Recommendations

- Focus more on high-attrition job roles, especially Sales Representatives.
- Review workload and overtime for employees who regularly work extra hours.
- Provide mentoring and training for younger and early-career employees.
- Give employees better career-growth opportunities.
- Improve job satisfaction through regular feedback, recognition and manager support.
- Improve work-life balance through better workload and shift planning.
- Review compensation and benefits for high-risk roles.
- Regularly check attrition by department, role, overtime, satisfaction and tenure.

11. Conclusions

The project gives a simple view of employee attrition and workforce patterns using Python and Power BI. The dashboard shows that attrition is different across departments, job roles and employee groups. Overtime, job satisfaction, work-life balance and shorter tenure are important areas to look at when studying retention. The results can help HR teams focus their attention on high-risk employee groups and improve retention activities.